

Natural Color QC Report

Intake Spectrophotometer batch — June 2024

Report date	21 June 2026
Lots measured	8
Ingredients	Grape Skin Anthocyanin · Beta-Carotene · Spirulina Phycocyanin
Instrument	Spectrophotometer D65 CIE 1931 2°
Colour standard	CIEDE2000 (dE ₀₀) Spectral RMSD
Workflow	color_workflow.py v1.0

3 PASS Released to formulation	2 MARGINAL Human review required	3 FAIL Quarantined
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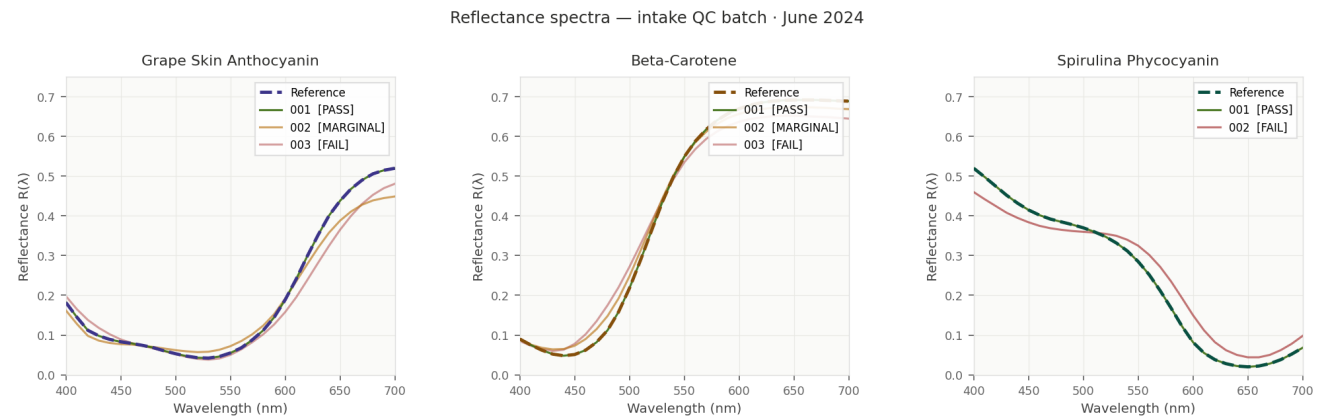
1 QC certificate — all lots

Lot ID	Ingredient	Supplier	L*	a*	b*	K/S	dE	RMSD	Decision
ANT-2024-001	Grape Skin Anthocyanin	Supplier-A	38.2	29.8	-12.4	10.926	0.0	0.0	PASS
ANT-2024-002	Grape Skin Anthocyanin	Supplier-A	41.5	24.2	-8.6	7.8	4.207	0.0314	MARGINAL
ANT-2024-003	Grape Skin Anthocyanin	Supplier-B	36.8	26.5	-15.2	12.177	2.793	0.0358	FAIL
BCR-2024-001	Beta-Carotene	Supplier-A	62.4	18.2	64.8	9.441	0.0	0.0	PASS
BCR-2024-002	Beta-Carotene	Supplier-A	63.8	14.5	68.2	6.844	3.062	0.0187	MARGINAL
BCR-2024-003	Beta-Carotene	Supplier-C	65.1	10.2	71.4	7.363	6.261	0.0353	FAIL
SPR-2024-001	Spirulina Phycocyanin	Supplier-C	48.6	-8.2	-32.5	24.01	0.0	0.0	PASS
SPR-2024-002	Spirulina Phycocyanin	Supplier-C	52.8	-4.5	-22.8	10.386	6.622	0.0399	FAIL

Tolerance gates: $dE \leq 2.0$ (CIEDE2000) | Spectral RMSD: ANT ≤ 0.025 , BCR ≤ 0.020 , SPR ≤ 0.030 . MARGINAL = within 130% of tolerance.

2 Reflectance spectra $R(\lambda)$

Each panel compares all intake lots for one ingredient against the approved reference spectrum (dashed). Deviation in curve shape — not just offset — indicates degradation or supplier-grade substitution.



3 Colour deviation ΔE per lot

CIEDE2000 colour difference vs. the approved reference. The dashed vertical line marks the 2.0-unit tolerance — the boundary of perceptible difference for a trained observer under D65.

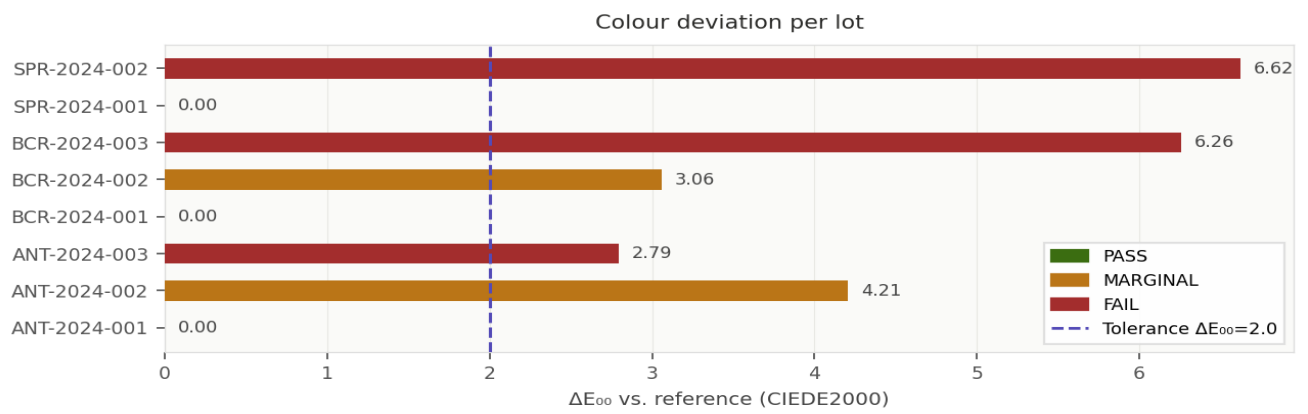


Figure 2 — CIEDE2000 ΔE_{00} . Green=PASS, amber=MARGINAL, red=FAIL.

4 Spectral overlay — measured vs. K-M correction prediction

For each non-passing lot, the Kubelka-Munk mixing model predicts the post-correction reflectance curve (green dotted). The shaded region shows the current spectral gap vs. reference. When the predicted curve converges with the reference, the correction recipe is reliable. Where it does not, the shape shift indicates degradation that pigment addition cannot fix.

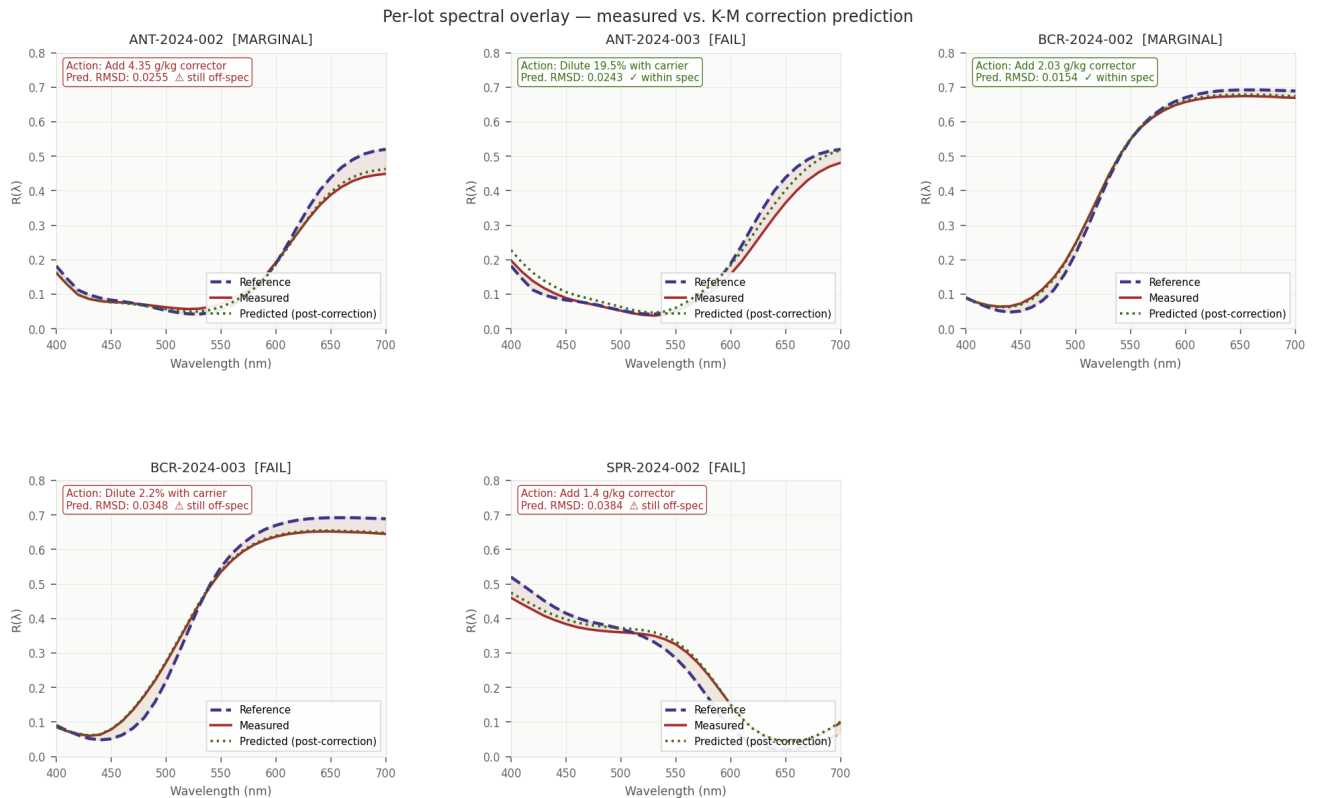


Figure 3 — Per-lot $R(\lambda)$ overlay. Dashed purple = reference. Red solid = measured. Green dotted = K-M predicted post-correction. Annotation shows action and predicted RMSD.

Reading the overlays: A green-dotted curve that tracks the reference closely means the corrective dose will work. A curve that remains offset (SPR-2024-002, BCR-2024-003) signals that the spectral shape — not just intensity — has changed, pointing to pigment degradation or a different grade of raw material.

5 Lot-level action summary

ANT-2024-001	PASS	Released to formulation engine. No action required.
ANT-2024-002	MARGINAL	Under-concentrated. Add 4.35 g/kg anthocyanin concentrate. K-M model predicts post-correction RMSD within spec. Recheck before release.
ANT-2024-003	FAIL	Over-concentrated (K/S 12.18 > ref 9.80, Supplier B). Dilute 20% with approved carrier or quarantine. Verify pigment grade.
BCR-2024-001	PASS	Released to formulation engine. No action required.
BCR-2024-002	FAIL	Under-concentrated. Add 2.03 g/kg Beta-Carotene 30% dispersion. Post-correction RMSD predicted within spec. Recheck before release.
BCR-2024-003	FAIL	Over-concentrated and hue-shifted ($dE_{ab}^* = 6.26$, Supplier C). Quarantine. Stability model flags 454 d predicted shelf life (<500 d minimum). Raise supplier issue; do not use in shelf-life-critical formulations.
SPR-2024-001	PASS	Released to formulation engine. No action required.
SPR-2024-002	FAIL	Spectral shape collapse (RMSD 0.040). K/S drop from 24.01 to 10.39 indicates phycocyanin degradation. Corrective addition insufficient — K-M predicted curve remains off-spec. Quarantine and investigate cold-chain.

6 Stability model — shelf-life predictions

Gradient-boosted regression trained on 5 lots with confirmed shelf-life data. CV RMSE: 127 days. Water activity and processing temperature account for 80% of predictive importance.

Lot ID	Ingredient	Predicted	Min spec	Status
ANT-2024-003	Grape Skin Anthocyanin	384 days	320 days	Above spec
BCR-2024-003	Beta-Carotene	454 days	500 days	BELOW SPEC
SPR-2024-002	Spirulina Phycocyanin	295 days	270 days	Above spec

BCR-2024-003 predicted at 454 d — 46 d below minimum. Send for accelerated shelf-life testing (40°C/75% RH) before any use. Feed confirmed results back to retrain the model.

7 Recommended next steps

ANT-2024-002 (MARGINAL)

Apply correction recipe (4.35 g/kg). Re-measure and resubmit through Steps 3–5 before release.

ANT-2024-003 (FAIL — over-conc.)

Dilute 20% with approved carrier or reject. Verify Supplier B grade spec.

BCR-2024-002 (FAIL — under-conc.)

Apply correction recipe (2.03 g/kg). Re-measure before release.

BCR-2024-003 (FAIL — shift + stability)

Quarantine. Raise supplier issue with Supplier C. Do not use in shelf-life-critical formulations.

SPR-2024-002 (FAIL — degraded)

Investigate cold-chain compliance. Request CoA from Supplier C. If breach confirmed, reject and re-order.

Model improvement

Send ANT-003, BCR-003, SPR-002 for ASLT (40°C/75% RH). Feed confirmed shelf-life back to stability model to reduce CV RMSE.